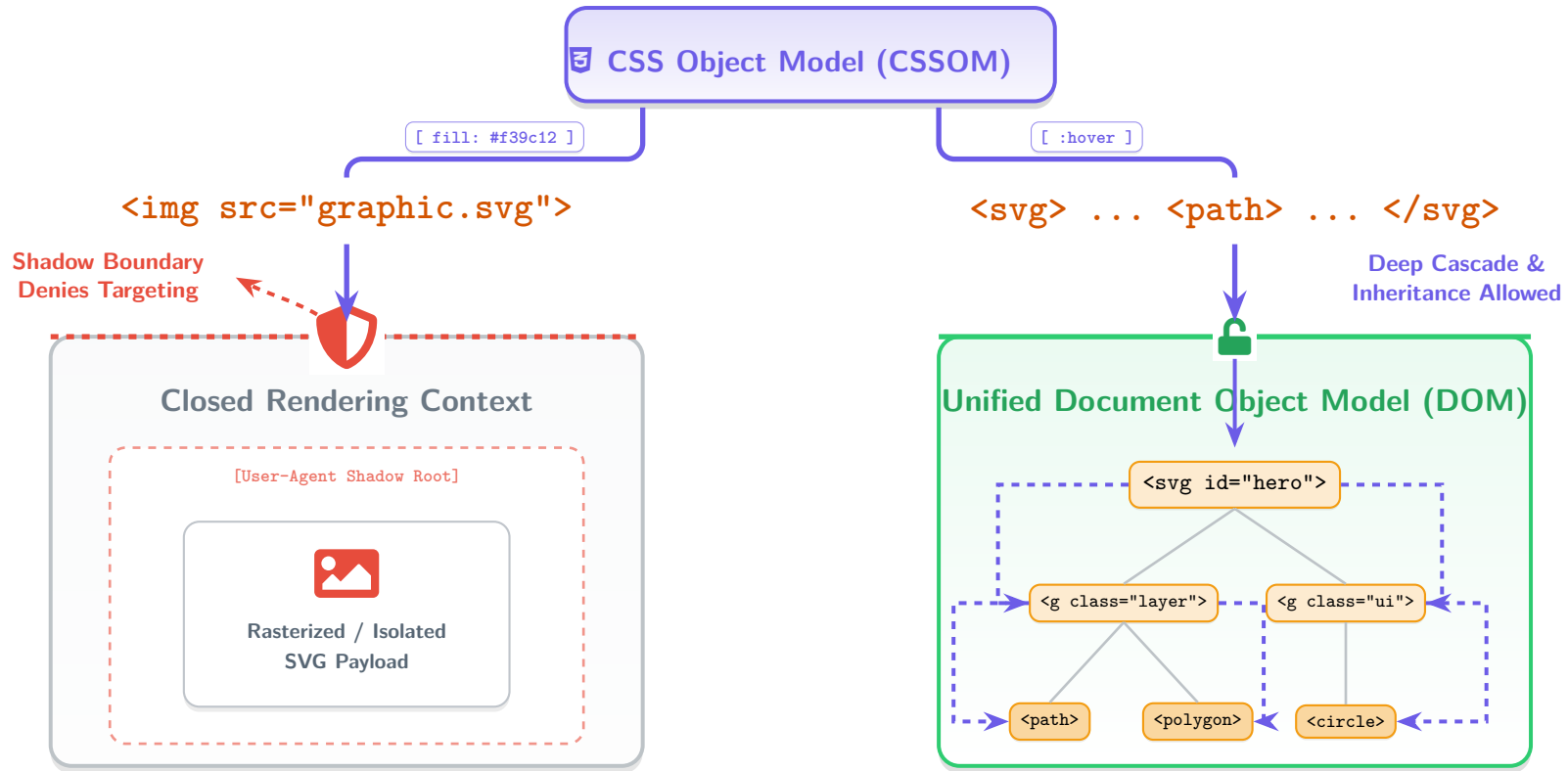


Inline SVG vs. External Reference: DOM Integration

Understanding the Topological Barrier to CSS Styling & The CSSOM



The Isolation Boundary

When SVG is loaded via an `<img>` tag or CSS `background-image`, the browser instantiates it within an isolated rendering sandbox.

Result: The master document's CSS Object Model cannot traverse this boundary. Complex interactivity, state changes (`:hover`), and specific geometric fills are locked inside a monolithic black box.

DOM Injection & The Topography

By parsing the SVG markup directly "inline" into the HTML document, the parser translates every mathematical node into a first-class DOM element (e.g., `SVGSVGElement`, `SVGPathElement`).

Result: A unified topological tree is created. The CSS cascade flows naturally into the vector geometry, allowing us to target individual paths, vertices, and groups just like standard `<div>` elements.